Federated Retrieval-Augmentation: Search as Individuation, Answers as Societies of Agents, and Why a True Answer Can Be a Useless One

Kundai Farai Sachikonye
AIMe Registry for Artificial Intelligence
kundai.sachikonye@bitspark.com

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Abstract

A retrieval-augmented system that returns a passage whose tokens match the query, drawn from a source that is true, can still be useless to the person who asked. We give a formal account of why this failure is not a ranking defect to be tuned away but a category error forced by treating *content* as though it were *meaning*. We develop the account entirely in finite weighted graphs. A corpus, a knowledge base, and a querying agent are each a *contact graph*: a finite weighted graph with a distinguished vertex, the *medium*, standing for everything not yet individuated. From the single hypothesis that the medium is never exhausted by any finite retrieval process, we derive a strictly positive *resolution floor* on every result: no retrieval returns a point, only a bounded region, and *content*—a passage's literal tokens—and *meaning*—the passage's individuating cut against a specific querying agent's own graph—provably come apart. We prove a **recognition/search identity**: retrieving for a query and recognising a match are inverse readings of one relation, so there is no third, privileged place, neither in the corpus nor in the querier, where "the answer" resides; it exists only as the joint fact of a decode. We prove **receiver-relativity**: the same passage registers a different, equally correct cell in different querying agents, so a single global relevance score is not merely imprecise but is asking a question that has no receiver-independent answer. We then replace the retrieve-then-generate pipeline with a **society of split-attention agents**: each pipeline stage is a bounded agent with its own conserved identity, its own purpose as the attractor of a strongly convex drive, and a finite attention budget it divides by a provably optimal water-filling rule across concurrently competing information sources (*scenes*). Retrieval becomes **individuation, not lookup**: the causal propagation table $\mathcal{T}(v_0, x^\star, \Pi)$, evaluated at a trivial process, returns a static knowledge-graph catalogue; evaluated at a nontrivial process, it returns a generated answer; the two are one object at two settings of one argument. Coherence of a returned claim requires a strongly connected cycle of *at least three* independently sourced, mutually supporting retrieval catalysts, never a single passage and never two; and a search terminates by **closure**—no further available source can add a new answer class—never by a confidence threshold crossing, with contested closure (irreconcilable sources) reported as a first-class output rather than silently resolved by picking one. Cross-checking answers drawn from sources with no shared internal representation (a trained domain adapter against a symbolic index, or two independently fine-tuned models) is certified by a **four-column relaxation** to *semantic quiescence*, which audits the *route*—what each answer would provoke as a follow-up—rather than the surface content, catching the false-friend failure in which two sources agree lexically while diverging in what they individuate. We instantiate the architecture as a concrete revision of the Four-Sided Triangle pipeline for a bounded research-group deployment: internal documents, domain-adapted models, and structured records are federated receivers in one contact graph, queried by individuation against a specific researcher's own decoder rather than scored against a receiver-independent notion of relevance. We report a validation suite of reproducible, seeded computational experiments against every load-

bearing claim, stored as machine-readable JSON records.

Keywords: retrieval-augmented generation; individuation by negation; resolution floor; content versus meaning; receiver-relativity; causal propagation table; water-filling attention allocation; coherence triangle; closure; four-column relaxation; federated search; split-attention agents.

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Part I
Foundations: Why a True Result Can Be a Useless One

1 Introduction

1.1 The failure this paper explains

Consider the ordinary operation of a retrieval-augmented generation system built the standard way: a query is embedded, the embedding is compared against a vector index of chunked documents, the top-k chunks by cosine similarity are retrieved, and a language model is asked to answer using them. The system returns a passage. The passage is drawn from a genuine, peer-reviewed, factually accurate source. Its tokens substantially overlap the query's tokens, which is why the retriever ranked it first. And it does not answer the question the researcher asked, because the researcher's question was never really about those tokens; it was about a specific configuration of a specific problem in a specific project, for which the retrieved passage is a true statement about something adjacent, not a statement about that.

This is not a tuning failure correctable by a better embedding model, a reranker, or a larger retrieved-context window. We will argue, and prove in the formal development below, that it is the symptom of a category error: the retrieval step optimises *content fidelity*—how well the returned string's literal tokens match the query's—while what the researcher requires is *meaning fidelity*—whether the returned material individuates the same thing the researcher is individuating, relative to what that researcher already holds fixed. These are, in general, different targets, and a system tuned against the first can degrade arbitrarily against the second while its reported relevance scores improve. A true, well-sourced, high-similarity passage is exactly the object this paper explains as *useless*: correct content, mismatched meaning.

1.2 The library, restated formally

An intuition motivates the formal development and is worth stating before the machinery, though no theorem below depends on it. A search engine and its user resemble a person standing before the shelves of a library. Neither the books nor the person holds the answer prior to the encounter: a book does not know, of itself, whether it is the answer to a question that has not been asked of it, and a person does not know, in advance of consulting the shelves, exactly which book will resolve their question. The answer is not stored in either party; it is constituted by the act of a specific person consulting a specific shelf under a specific question, and a different person consulting the same shelf under the same words may correctly arrive elsewhere. Two complementary strands of a DNA molecule offer the same structural point in a different register: neither strand, read alone, is the genetic information; the strands are different *representations* related by a fixed complementarity rule, and the invariant that is transmitted is a property of the paired structure, not of either sequence in isolation. We will show, as theorems rather than as this analogy, that a retrieval result is structurally the same kind of object: never a fact residing in a document, never a fact residing in a query, but a joint individuation computed against a medium that belongs to neither.

1.3 What this paper contributes

1. **The resolution floor for retrieval** (Section 2). We derive, from the single hypothesis that the search medium is never exhausted, a strictly positive floor $\beta > 0$ on every retrieval: no result is a point, every result is a bounded region.
2. **Content and meaning provably diverge** (Section 3). We prove that a passage's content (its literal tokens) and its meaning (its individuating cut against a specific receiver's graph) coincide only in the degenerate case where the literal form is itself the operative individuation, and exhibit the general case in which a content-optimal retrieval is meaning-suboptimal by a quantity bounded away from zero.
3. **The recognition/search identity and receiver-relativity** (Section 4). We prove that querying and recognising a match are inverse readings of one relation, so there is no receiver-independent fact of relevance; distinct researchers correctly register distinct cells for the same passage, at their own floor, with no privileged registration among them.
4. **Retrieval as individuation via the causal propagation table** (Section 5). We show a static knowledge index and a generative retrieval-augmented answer are one table, $\mathcal{T}(v_0, x^\star, \Pi)$, evaluated at two settings of a process argument, unifying "look it up" and "reason it out" as the same operation.
5. **Coherence and closure as the correct stopping and grounding rules** (Section 6). We prove that a claim is robustly grounded only by a strongly connected triangle of independent sources, never one passage and never two, and that a search is correctly terminated by *closure*—no further available source adds a new answer class—which we prove strictly stronger than any confidence threshold.
6. **The four-column route-audit for cross-source verification** (Section 7). We construct a relaxation that certifies agreement between sources with no shared internal representation by comparing what each source's answer *provokes*, catching false friends invisible to endpoint-only comparison.
7. **The society-of-agents architecture** (Section 8). We replace the fixed sequential

retrieve-then-generate pipeline with a society of bounded, split-attention agents, each a node of one contact graph, coordinating via a shared outcome space rather than a shared representation, and dividing attention across concurrently competing sources by a provably optimal water-filling rule.

8. **A concrete instantiation for a bounded research-group deployment** (Section 9). We specify how the architecture federates internal documents, domain-adapted models, and structured records as receivers in one contact graph queried by individuation against a named researcher's own decoder.

9. **Validation** (Section 10). We report a seeded, reproducible computational suite checking every load-bearing theorem on explicitly constructed instances, with results stored as JSON records.

1.4 Method

Every object below is a finite weighted graph, a bounded map between finite sets, or a construction on one. Minimum cuts are exact and computable by maximum flow (Ford and Fulkerson, 1956; Menger, 1927). We introduce no probability measure and no metric beyond a derived normalisation on $[0, 1]$; where an uncertainty is required it is the graph's own floor-distance, forced by the construction rather than posited by a prior or a membership function. Words with an evocative reading—"content," "meaning," "relevance," "coherence"—each name a specific graph-theoretic object defined in the text; no theorem depends on any reading beyond that definition. Interpretive remarks are marked as such and are never load-bearing.

2 The Search Medium and Its Floor

2.1 Contact graphs for retrieval

Definition 2.1 (Contact graph; the medium)**.** A *contact graph* is a finite weighted graph $\Gamma = (V, E, w)$, $w : E \to \mathbb{R}_{>0}$, together with a distinguished vertex $\mathsf{m} \in V$, the *medium*, adjacent to every other vertex. Vertices $v \neq \mathsf{m}$ are *claims*: candidate individuated units of information—a document chunk, a database record, a structured fact, a model-generated statement. An edge $\{u, v\}$ is a *contact*; its weight is the cost of the distinction separating u from v. The medium stands for everything not yet individuated, without regard to whether it is stored on a local index, a remote corpus, or the weights of a trained model.

Remark 2.2 (No boundary between local, remote, and learned). Theorem 2.1 draws no distinction between a claim recoverable from a locally indexed document, one recoverable from a remote database, and one recoverable only by an inference from a trained model's parameters: all three are, prior to individuation, contents of the single vertex m. This licenses the federation of Section 8: a retrieval pipeline may draw candidate claims from a full-text index, a knowledge graph, and a domain-adapted language model as computationally uniform sources, because the graph encodes only a cost of separation, never a storage medium.

Definition 2.3 (Separation cost)**.** The *separation cost* of a claim v is $\sigma(v) = \min_{S \ni v,\, \mathsf{m} \notin S} w(\partial(S))$, attained since V is finite; the minimising cut is v's *resting cut*.

2.2 Non-completability and the floor theorem

Definition 2.4 (Non-completable medium)**.** An *expanding retrieval family* is a chain of contact graphs $\Gamma_0 \subseteq \Gamma_1 \subseteq \cdots$, each obtained from the last by adding claims and contacts while retaining the medium and all prior structure. The medium is *non-completable* if the family has no terminal stage: for every finite Γ_k there is $\Gamma_{k+1} \supsetneq \Gamma_k$. This is an order condition—no last stage of retrieval—not a claim about the cardinality of the corpus.

Remark 2.5 (Why a retrieval medium is non-completable). For any finite index—however large the corpus, however well the model was trained—there is a further document not yet ingested, a further record not yet linked, a further inference not yet drawn. This is the formal content of the observation that a research group's internal corpus, once federated with external literature and a trained model's own generative capacity, has no boundary at which retrieval could in principle be finished.

Theorem 2.6 (Floor from non-completability)**.** *Let the medium be non-completable and let v be a claim ($v \neq \mathsf{m}$, v not exhausting V). Then no finite retrieval stage completes v's identification*

against the whole medium, and the identification residual $\rho(v) > 0$.

Proof. Suppose $\rho(v) = 0$: identification is complete at some finite Γ_k, i.e. every claim of the medium has entered the computation of v's cut, so Γ_k contains the whole medium. This contradicts Theorem 2.4, which supplies $\Gamma_{k+1} \supsetneq \Gamma_k$ containing a claim absent from Γ_k and hence not yet brought to bear. No finite stage completes the identification, so $\rho(v) > 0$. □

Theorem 2.7 (Resolution floor)**.** *There is* $\beta > 0$ *with* $w(e) \geq \beta$ *for every contact and* $\sigma(v) \geq \beta$ *for every claim: no retrieval separates a claim from the medium at zero cost, and no result is returned for free.*

Proof. Set $\beta := \inf_v \rho(v)$ over the finitely many claims, each positive by Theorem 2.6; an infimum of finitely many positive reals is positive. The residual is carried by the edges of the resting cut, so every such edge has weight $\geq \beta$; every contact lies in some singleton's cut, so every edge weight is $\geq \beta$; connectivity forces non-empty cuts, so $\sigma(v) \geq \beta > 0$. □

Corollary 2.8 (No exact result)**.** *No retrieval returns a claim separated from the medium by a cut of weight zero. A system that reports a result as "exact" has, in fact, resolved to a region of width bounded below by* β.

Remark 2.9 (The floor is not an engineering defect). Theorem 2.7 does not say retrieval is imprecise because indexes are imperfect. It says retrieval against a genuinely inexhaustible medium cannot, even with a perfect instrument, terminate at zero residual, because that would require completing a comparison against a medium that, by hypothesis, is never completed. Section 6 makes this the basis of an honest stopping rule, closure, rather than a caveat to apologise for.

3 Content and Meaning Provably Diverge

We now state, in the vocabulary of Section 2, the structural fact behind the failure of Section 1: what a passage *says*, taken as a string, and what it *individuates* for a specific receiver, are different objects, coincide only in a degenerate case, and a retriever tuned to the first can be arbitrarily wrong about the second.

Definition 3.1 (Content versus meaning)**.** The *content* of a claim v is its label: the literal tokens of the passage, record, or statement it names. The *meaning* of v relative to a receiver graph Γ_R is v's resting cut in Γ_R (Theorem 2.3): the region of contrasts individuating v against that receiver's own rest. Content is a name on the claim; meaning is the claim's individuation against a specific whole. They coincide only when the literal form is itself the operative individuation in Γ_R, which Theorem 3.3 shows is not the general case.

Example 3.2 (The true, irrelevant passage)**.** A researcher investigating why a specific enzyme assay in their own pipeline fails under a specific buffer condition queries a federated index. The top-ranked passage by token overlap is a genuine, correctly cited textbook statement of the enzyme's general kinetics: true, sourced, lexically close to the query. Its content is exactly what a content-similarity retriever is built to find. Its meaning relative to the researcher's own graph—which already holds fixed the specific assay, the specific buffer, and the specific observed failure—is the cell *textbook background*, not the cell *explanation of this failure*; the passage individuates against *other general kinetics statements*, not against *other candidate causes of this specific failure*. The two individuations are different regions of two different graphs, and no increase in token-similarity ranking moves the first toward the second.

Theorem 3.3 (Content and meaning come apart)**.** *Let* Γ_C *(the corpus graph, carved by lexical and topical similarity) and* Γ_R *(a specific researcher's receiver graph, carved by the distinctions already fixed in their project) be two contact graphs over claims of a shared domain, with different edge weightings. A content map* $c : V \to V$ *sending each claim to its highest-similarity correspondent does not in general preserve meaning: there exist claims whose content-correspondent has a resting cut in* Γ_R *different from the claim's own meaning in* Γ_R*. Two independent obstructions guarantee this:* (i) different carvings—Γ_C *may resolve one cell where* Γ_R *resolves several (or the reverse), so a content-nearest claim in* Γ_C *need not land in the corresponding cell of* Γ_R*;* (ii) content is not meaning even within one carving—*the correspondent's content is preserved by construction, but content is, by Theorem 3.1, exactly what meaning is not.*

Proof. (i) Let Γ_C resolve a single cell $\{v\}$ where Γ_R resolves two cells $\{v_1\}, \{v_2\}$ under a finer partition of the same underlying claims induced by the researcher's already-fixed distinctions. The content map sends a query to v; v's resting cut in Γ_R individuates against whichever of v_1, v_2 it is *not*, a contrast absent from v's undivided cell in Γ_C. The two cuts differ. (ii) By Theorem 3.1, a content correspondent is selected by similarity of label; its resting cut in Γ_R is a property of Γ_R's structure, into which label similarity does not enter as a defining term. In Theorem 3.2 the correspondent maximises lexical overlap (content) while its resting cut individuates against *other general statements* rather than against *other candidate explanations of the specific failure* (meaning); the two differ. □

Theorem 3.4 (Meaning is unextractable, so it cannot be read off by a scorer)**.** *There is no operation that reads a claim's meaning off as a residue-free value, for any receiver graph Γ_R. Such a reading would require $\rho(v) = 0$ in Γ_R, which by Theorem 2.6 requires Γ_R's whole medium to be brought to bear on v—i.e. the scorer must occupy the entirety of v's negation structure in Γ_R, which is not a comparison but an identity.*

Proof. By Theorem 2.7, v's meaning is a cut of weight $\geq \beta > 0$ in Γ_R; reducing it to a residue-free point is $\rho(v) = 0$, excluded by Theorem 2.6. A relevance scorer computing a single real number from v's content and the query's content, without access to Γ_R's full structure, is precisely attempting this reduction; by the argument above it cannot succeed for every receiver, since Γ_R varies across receivers (Theorem 4.8 below) and is not, in general, recoverable from content alone. □

Corollary 3.5 (A content-similarity score is not a meaning score)**.** *A retrieval system that ranks and returns claims by a function of query content and claim content alone—cosine similarity of embeddings, BM25, or any learned reranker trained to imitate human relevance labels attached to content pairs—computes an approximation to content correspondence (Theorem 3.3), never to meaning (Theorem 3.4), and can be improved on the first axis while remaining arbitrarily wrong on the second. This is the precise, provable content behind the failure of Section 1: the system was never computing the quantity the researcher needed, however well it was tuned to compute the quantity it was built to compute.*

4 Individuation by Negation and Receiver-Relativity

We now show why there is no fix available within the content-scoring paradigm: relevance is not merely hard to compute, it is not a receiver-independent fact.

4.1 Individuation by negation

Definition 4.1 (Selector; no privileged claim)**.** A *selector* for a claim set $U \subseteq V$ is a rule fixing U by naming a distinguished member in advance. A determination is *selector-free* if it names no member of U. The graph has *no privileged claim* if the data available to a determining rule are invariant under the weighted automorphism group $\mathrm{Aut}(\Gamma)$.

Theorem 4.2 (Individuation by negation)**.** *In a medium with no privileged claim, any selector-free determination of a claim set U fixes it as $U = V \setminus \complement U$ with the negation set $\complement U$ specified first; conversely every admissible $\complement U$ determines a unique U. No selector-based rule fixes U without invoking a further selector.*

Proof. Necessity. A positive rule naming a distinguished member of U must, since the data are $\mathrm{Aut}(\Gamma)$-invariant, itself supply that member as a selector; that selector is either given in advance (excluded) or supplied by a prior rule facing the same regress. The regress terminates only at a rule naming no member of U: specifying $\complement U$ and taking U as the remainder. *Sufficiency.* Complementation is an involution on subsets of the finite V, so $\complement U$ determines U uniquely. □

Corollary 4.3 (A claim is fixed by what it contacts and what it does not)**.** *A single claim v is determined selector-free by the partition of the rest into $N(v)$ (its contacts) and $V \setminus N[v]$ (neither v nor its contacts). No intrinsic label distinguishes v; its identity is carried by this negation structure relative to the ambient graph.*

4.2 The recognition/search identity

Definition 4.4 (Query; decoder; projector)**.** A *query* is any string or structured value presented to the search process. A *decoder* is $\mathrm{Dec} : Q \to V$

sending a query to the claim it individuates. A *projector* is $\mathrm{Proj} : V \to 2^Q \setminus \{\varnothing\}$, $\mathrm{Proj}(v) := \mathrm{Dec}^{-1}(\{v\})$.

Theorem 4.5 (Recognition/search identity)**.** *For every claim v and query q:* $\mathrm{Dec}(q) = v \iff q \in \mathrm{Proj}(v)$. Recognition—*computing* $\mathrm{Dec}(q)$ *for a percept—and* search—*computing a representative of* $\mathrm{Proj}(v)$ *for a target claim—are inverse readings of the single relation* $\{(q, v) : \mathrm{Dec}(q) = v\}$*; neither is definable without the other.*

Proof. Immediate from $\mathrm{Proj}(v) = \mathrm{Dec}^{-1}(\{v\})$: membership $q \in \mathrm{Proj}(v)$ is definitionally equivalent to $\mathrm{Dec}(q) = v$. A decoder and its fibre map determine each other on a finite domain: Dec is recovered from $\{\mathrm{Proj}(v)\}_v$ as the map sending each q to the unique v with $q \in \mathrm{Proj}(v)$, and Proj is recovered from Dec by Theorem 4.4. □

Remark 4.6 (Neither the corpus nor the querier holds the answer). Theorem 4.5 is the formal content of the library intuition of Section 1. The document does not, of itself, hold the answer to a question not yet posed to it: it is one term of a relation whose other term is the query. The querier does not, of itself, hold the answer prior to consulting the medium: a query is itself an individuation, exactly as exact as recognition ever is, not a degraded version of a specification the querier secretly already possesses. The answer is the joint fact $\mathrm{Dec}(q) = v$, and it exists only at the encounter of the two.

4.3 Receiver-relativity: no privileged relevance

Definition 4.7 (Receiver; registered cell)**.** A *receiver* is a contact graph, understood as a specific querying agent's own decoder structure—the distinctions that agent already holds fixed. The *cell* a claim v *registers* in a receiver is v's resting cut computed in that receiver's graph.

Theorem 4.8 (Distinct receivers register distinct, equally correct cells)**.** *The same claim registers different cells in receivers with different decoder graphs, and each registration is a correct individuation at that receiver's own floor. No registration is privileged as "the" relevance of the claim, and there is no receiver-independent relevance value of which the registrations are approximations.*

Proof. By Theorem 4.7 the registered cell is the resting cut in the receiver's own graph; different graphs, carrying different already-fixed distinctions, yield different minimum cuts for the same claim (a sparse receiver graph individuates the claim against few contrasts, a dense one against many). Each is the minimum-cut individuation in its own graph and so attains that graph's floor (Theorem 2.7), hence is correct there. Were a privileged relevance value to exist, a registration differing from it would be an error; but every registration is correct in its own graph, and by Theorem 3.4 no receiver-independent value exists to be approximated. Hence none is privileged. □

Corollary 4.9 (A single global relevance score answers no receiver's question)**.** *A retrieval system that computes one relevance score per (query, claim) pair, independent of which researcher's already-fixed context the query is issued from, is computing an object Theorem 4.8 shows does not exist as a receiver-independent fact. It is not merely coarse; it targets a quantity with no referent.*

Remark 4.10 (The two researchers). A junior researcher new to a project and the project's principal investigator, issuing the identical query string against the identical federated index, are different receivers in the sense of Theorem 4.7: the junior researcher's graph holds few prior distinctions fixed, so a broad textbook passage may correctly register as the useful cell; the principal investigator's graph holds the entire project history fixed, so the same passage correctly registers as *already known, uninformative*, and a narrower, more technical claim registers as the useful one. Neither is wrong. Section 9 turns this into a design requirement: retrieval must be individuated against a specific researcher's accumulated graph, not scored once against the query alone.

Part II
Retrieval as Individuation

5 The Causal Propagation Table: One Object, Two Readings

We now unify "look it up" and "reason it out" as one operation, removing the architectural distinction between a static knowledge index and a generative retrieval-augmented answer.

Definition 5.1 (Alignment)**.** Fix a contact graph and target claim $x^\star$. The *alignment* of x to $x^\star$ is $\sigma(x, x^\star) = \min_{S \ni x,\, x^\star \notin S} w(\partial(S)) \in [\beta, \Omega]$, $\Omega := w(E)$; the *alignment score* is $a(x, x^\star) := \sigma(x, x^\star)/\Omega \in (0, 1]$.

Definition 5.2 (Process graph; propagation; accountability)**.** A *process graph* is $\Pi = (V, E, w; I, x^\star)$ with seeds $I \subseteq V$ and target $x^\star \in V$. A *propagation* of a seed $v_0 \in I$ is a walk $W = (v_0, e_1, \ldots, e_k, v_k)$ with $v_k = x^\star$. The *trivial process* $\Pi_{\text{rest}}(v)$ has $I = \{v\}$, $x^\star = v$, and single operation $\{v, \mathsf{m}\}$. Fix a tolerance $\varepsilon \geq 0$ and system floor $\beta^\star := \min_v \sigma(v)$; a propagation is *$\varepsilon$-accountable* if $\sigma(v_0, x^\star) \leq \beta^\star + \varepsilon\,\Omega$.

Definition 5.3 (Causal propagation table)**.**

$$\mathcal{T}(v_0, x^\star, \Pi) := \begin{cases} \partial(S^*(x^\star)) & \text{if some propagation is accountable,} \\ \varnothing & \text{otherwise.} \end{cases}$$

Theorem 5.4 (The directional identity: catalogue and generation are one table)**.** *For every claim v: $\partial(S^*(v)) = \mathcal{T}(v, v, \Pi_{\text{rest}}(v))$, so the resting catalogue of the graph—the assignment $v \mapsto \partial(S^*(v))$ that a static knowledge index computes—is exactly the table restricted to trivial processes. For a nontrivial process Π, $\mathcal{T}(v_0, x^\star, \Pi)$ is an object of the* same type*: an accountable cut of weight $\geq \beta$, the convergent terminus of a propagation, computed under process rather than at rest.*

Proof. Taking $\Pi_{\text{rest}}(v)$, the sole propagation is the one cut event individuating v against m; it is admissible since it terminates at its own target, and its accountable terminal object is $\partial(S^*(v))$ by Theorem 5.3, giving the forward equality. For a nontrivial Π, every admissible propagation terminates at $x^\star$, whose individuating datum is the resting cut $\partial(S^*(x^\star))$ of weight $\geq \beta$ (Theorem 2.7), the same type of object as a catalogue cell, differing only in being the terminus of a nontrivial walk rather than of the rest contact. □

Remark 5.5 (No architectural seam between retrieval and generation)*.* Theorem 5.4 removes a design distinction the current retrieve-then-generate pipeline treats as fundamental. A knowledge graph lookup and a multi-hop generative answer are not two subsystems requiring an interface; they are the identical table evaluated at two settings of one parameter. A system built on this identity has one execution path, parameterised by process complexity, rather than a retrieval component bolted to a generation component.

Theorem 5.6 (Convergence admissibility and path opacity)**.** *A propagation is admissible iff its terminal claim is $x^\star$; interior claims are entirely unconstrained and may have arbitrarily large alignment to $x^\star$ without invalidating the propagation. No invariant computed from $(v_0, x^\star)$ alone distinguishes two admissible propagations differing only in their interior.*

Proof. Admissibility depends, by Theorem 5.2, only on the terminal condition $v_k = x^\star$; nothing constrains interior claims. Any invariant computed from the endpoints alone is by hypothesis identical on two propagations sharing endpoints, and cannot recover interior data not determined by those endpoints. □

Remark 5.7 (A strange intermediate step is not evidence of failure)*.* Theorem 5.6 licenses a multi-hop retrieval-and-reasoning trajectory to pass through an intermediate claim that looks unrelated to the eventual answer, without this being evidence the search has gone wrong. Evidence of failure is non-convergence—the walk never reaching a claim in the target region—never the appearance of any single intermediate step. A production system should log and permit inspection of intermediate steps for auditability, but must not gate continuation on their surface plausibility.

6 Catalytic Composition, Coherence, and Closure

6.1 Catalysts and the multiplicative law

Definition 6.1 (Catalyst; catalytic power)**.** A *catalyst* toward $x^\star$ is a partial map $\gamma : V \to V$ realising a contact operation. Its *power* at x is $\kappa(\gamma, x, x^\star) = \dfrac{a(x, x^\star) - a(\gamma(x), x^\star)}{a(x, x^\star) - \beta/\Omega} \in [0, 1]$.

Theorem 6.2 (Multiplicative composition)**.** *For catalysts $\gamma_1, \ldots, \gamma_n$ applied in sequence with powers $\kappa_1, \ldots, \kappa_n$, the composite power is $\kappa(\gamma_1 \cdots \gamma_n) = 1 - \prod_i (1 - \kappa_i)$.*

Proof. Let r_0 be the initial above-floor gap. Each application leaves $r_i = (1 - \kappa_i) r_{i-1}$; by induction $r_n = \prod_i (1 - \kappa_i) r_0$, and the fraction closed is $1 - r_n / r_0$. □

Corollary 6.3 (Diversify sources, do not repeat them)**.** *Repeating one source n times yields power $1 - (1 - \kappa)^n$, strictly increasing but never reaching 1; a chain of near-duplicate retrieval calls to one index accrues power slowly and may fail to saturate even in principle, while independent catalysts addressing distinct evidentiary channels (a full-text index, a structured record store, a domain-adapted model's own inference) saturate faster and more robustly. This is forced by the multiplicative law, not an empirical heuristic about retrieval engines.*

6.2 Coherence requires a triangle

Definition 6.4 (Support graph)**.** Catalysts γ_j *supports* γ_i toward common target $x^\star$ at threshold $\theta \in (0, 1]$ if γ_j's result reinforces, to degree $\geq \theta$, the alignment shift γ_i induces. The *support graph* has edge $j \to i$ under this relation.

Theorem 6.5 (Linear chains and pairs fail; a triangle grounds)**.** (i) *No acyclic chain of catalysts, in which the terminal member is justified by nothing internal to the chain, is robust to removal of that terminal member.* (ii) *A 2-cycle—two catalysts justifying only each other, with no independent third check—fails the majority condition $\theta > \frac{1}{2}$: removing either leaves the other unsupported.* (iii) *Three catalysts mutually supporting one another at $\theta > \frac{1}{2}$, forming a strongly connected triangle, ground the claim robustly: their combined shift toward $x^\star$ survives the removal of any single member.*

Proof. (i) An acyclic support graph has a source with no incoming support edge; its removal (or the removal of anything supporting only acyclically downstream members) collapses everything below it, with no internal check to prevent this. (ii) A 2-cycle is mutual definition with no independent third opinion; removing either of the two catalysts leaves exactly one, itself unsupported, so no majority above $\theta > \frac{1}{2}$ can be formed from a single remaining voter. (iii) In a strongly connected triangle each catalyst is supported by the other two; at $\theta > \frac{1}{2}$ the two supporters jointly outweigh any single dissent, and removing one catalyst leaves the remaining two still mutually supporting each other directly, at a reduced but still positive effective threshold. □

Remark 6.6 (One passage is never a grounded answer, and neither are two). Theorem 6.5 is the precise, provable version of a longstanding but usually informal caution against single-source retrieval-augmented answers. A system returning one top-ranked passage, however high its content-similarity score, has a support graph of one vertex and no cycle; a system requiring agreement between exactly two sources has a support graph whose only possible cycle has length two, proved insufficient by Theorem 6.5(ii). Robust grounding is a structural minimum of three independently sourced, mutually reinforcing catalysts, decidable from the *signs* of pairwise support—whether each pair reinforces or undercuts the other—with no access to numerical confidence magnitudes.

6.3 Closure: the correct stopping rule

Definition 6.7 (Reachable class set; closure)**.** Let $\mathrm{cell}(v_0, \Pi)$ be the set of target claims reachable by an admissible propagation of v_0 through catalysts invoked so far, partitioned into classes under endpoint-indistinguishability (Theorem 5.6). A search is *closed* if, for every available but not-yet-invoked catalyst, extending the search cannot add a new class to $\mathrm{cell}(v_0, \Pi)$.

Theorem 6.8 (Closure is strictly stronger than a confidence threshold)**.** *There exist process graphs and thresholds θ such that a propagation's terminal alignment satisfies $a(x^\star, x^\star) \geq \theta$ while the search is not closed: a further, uninvoked catalyst reaches a target in a distinct class.*

Proof. Construct two disjoint, internally dense claim clusters A, B, each connected to the

medium by one catalyst of equal weight. A propagation via γ_A terminates at $x^\star_A \in A$ with terminal alignment at the floor, trivially satisfying any $\theta \leq 1$ (a propagation's terminal alignment to itself is always at the floor). Yet γ_B, not yet invoked, reaches $x^\star_B \in B$, a distinct class by construction. □

Theorem 6.9 (Convergent closure or honest decline)**.** *Over a finite catalyst registry, every search terminates in exactly one of:* (i) convergent closure*,* $\mathrm{cell}(v_0, \Pi)$ *stabilises to a single class, reported as the result;* (ii) contested closure*,* $\mathrm{cell}(v_0, \Pi)$ *stabilises with more than one class, reported as* decline *together with the distinct classes found. No search over a finite registry runs forever without reaching one of these.*

Proof. The catalyst registry is finite, so the reachable-target set is finite and $\mathrm{cell}(v_0, \Pi)$ can grow only finitely many times before every remaining catalyst has been tried against every currently-reached class, at which point Theorem 6.7 is satisfied by exhaustion. If the stabilised partition has one class, (i) holds; if more than one, no single sufficient claim is licensed and the search reports (ii). These exhaust the possibilities on a finite, nonempty partition. □

Remark 6.10 (Decline is a correct output, not an error). Theorem 6.9(ii) requires a retrieval system to report "the sources disagree, here are the distinct positions" as a first-class, correctly-terminating outcome, rather than silently emitting one contested class as though it were unique because it happened to rank highest by content similarity. For a research-group deployment this is not a fallback behaviour to minimise; it is frequently the more valuable output, since contested closure identifies exactly where the internal record and the published literature, or two internal documents, diverge.

7 Cross-Source Verification Without a Shared Representation

A federated deployment draws claims from sources with no shared internal representation: a domain-adapted language model's weights share no decoding with a structured database's schema. We construct a verification method that certifies agreement between such sources without requiring either to expose its internals, and that catches a failure mode invisible to ordinary relevance comparison.

Definition 7.1 (Opaque pair; column; cross-demand)**.** Two receivers $\mathcal{R}_A, \mathcal{R}_B$ are an *opaque pair* if no computable map relates their internal knowledge states. Given a common query, the *column* of $\mathcal{R}_A$ is its candidate output a; likewise b for $\mathcal{R}_B$. The *cross-demand* $D_{A\to B}$ is the above-floor alignment gap between b and the reverse-image of a in $\mathcal{R}_B$'s space; symmetrically $D_{B\to A}$.

Definition 7.2 (Provoked column; four-column construction)**.** The *provoked column* $\rho(a)$ of a column a is the receiver's own candidate output on the follow-up query a itself would generate under the receiver's internal policy (e.g. "what would need to be true for this to be wrong"), computed using only that receiver's own machinery. The *four-column system* comprises the central columns a, b and the provoked columns $\rho(a), \rho(b)$.

Definition 7.3 (Quiescence)**.** The relaxation alternating updates toward mutual consistency is *quiescent* at tolerance $\tau := \beta(\mathcal{R}_A) + \beta(\mathcal{R}_B)$ if all four pairwise cross-demands among $a, b, \rho(a), \rho(b)$ fall below τ and remain there.

Theorem 7.4 (The four-column criterion)**.** *The four-column system is quiescent iff* (a) a, b *are quiescent (same meaning) and* (b) $\rho(a), \rho(b)$ *are quiescent (same invited follow-up). Quiescence certifies identity of meaning by the* route*, not merely the endpoints, computed without either receiver disclosing its internal state.*

Proof. Quiescence of the system is simultaneous vanishing of all four cross-demands, which is exactly (a) and (b); each pairwise vanishing is identity of meaning of that pair (the dichotomy of Theorem 7.5 below), computed from cross-demands—minimum-cut weights over outputs alone, never from any decoded internal value. □

Theorem 7.5 (Dichotomy)**.** *The relaxation either reaches quiescence in finitely many rounds, or the residual sequence is bounded away from zero for all sufficiently large rounds. There is no third outcome on a finite candidate space.*

Proof. Each round strictly decreases the maximal residual unless already below tolerance, on a finite space of column values; such a sequence either reaches the tolerance band in finitely many

steps or its decrements vanish while the value remains above tolerance. These exhaust the possibilities. □

Theorem 7.6 (Route-audit: the four-column system catches a false friend)**.** *There exist opaque pairs and queries for which the central cross-demands $D_{A\to B}, D_{B\to A}$ are both below tolerance while the provoked cross-demands are not, so the pair is non-quiescent although an endpoint-only comparison would report agreement.*

Construction. Let both receivers answer a query with numerically identical central columns, so central cross-demands vanish trivially. Let $\mathcal{R}_A$'s answer provoke, under its own follow-up policy, a question presupposing condition c_1, and $\mathcal{R}_B$'s numerically identical answer provoke a question presupposing a non-equivalent condition c_2. The provoked answers concern different conditions, so their cross-demand need not be small although the central cross-demand is zero. □

Remark 7.7 (Two internal documents can agree for different reasons). Theorem 7.6 formalises a failure that occurs concretely in a research group's internal corpus: two documents—a protocol write-up and a results summary—may state numerically identical conclusions while one holds under an assumption the other's context silently violates. A retrieval system comparing only the two documents' stated content reports agreement; the four-column route-audit, by comparing what each document's claim would provoke as a follow-up question, exposes the divergence. This is the graph-theoretic content of what colloquially would be called verifying two sources actually mean the same thing, not merely that they say the same words.

Part III
Architecture: A Society of Split-Attention Agents

8 Federated Receivers and the Society Graph

We now assemble Section 2–Section 7 into an architecture and specify how it replaces a fixed, sequential retrieve-then-generate pipeline.

8.1 Bounded receivers and the federation floor

Definition 8.1 (Bounded receiver; federation)**.** A *receiver* is a triple $\mathcal{R} = (\mathcal{K}, \Phi, \Pi)$ on an outcome space $(\mathcal{X}, d, \Omega)$: $\mathcal{K}$ a finite internal state set, $\Phi : \mathcal{X} \to \mathcal{K}$ a decoder, $\Pi : \mathcal{K} \to 2^{\mathcal{X}} \setminus \{\varnothing\}$ a candidate projection. Its *floor* is $\beta(\mathcal{R}) = \sup_x \inf_{x' \in \Pi(\Phi(x))} d(x, x')$. A *federation* of receivers $\{\mathcal{R}_i\}$ has joint decoder and *union* candidate projection $\Pi^\dagger(x) = \bigcup_i \Pi_i(\Phi_i(x))$.

Theorem 8.2 (Federation lowers the floor)**.** $\beta(\mathsf{F}) \le \min_i \beta(\mathcal{R}_i)$ *for a federation* F*, strictly whenever non-redundant at every member; under independent resolution failure,* $1 - \beta(\mathsf{F})/\Omega = \prod_i (1 - \beta(\mathcal{R}_i)/\Omega)$.

Proof. For every x, $\Pi_j(\Phi_j(x)) \subseteq \Pi^\dagger(\Phi^\dagger(x))$ for each j, so the infimum over the (larger) union is at most the infimum over any single member's projection, hence at most the pointwise minimum over members; the supremum of a pointwise minimum is bounded above by the minimum of the suprema, giving $\beta(\mathsf{F}) \le \min_i \beta(\mathcal{R}_i)$, strict under non-redundancy by the standard argument. The multiplicative law follows from the independence hypothesis applied to the failure event at each floor. □

Remark 8.3 (Federation, not ensembling by vote). Theorem 8.2 requires the *union* construction: every receiver's candidate is offered as admissible, so a query on which one federation member (a domain-adapted model, say) is precise benefits the whole federation even where the others are not. This is architecturally distinct from an ensemble that averages or votes among receivers' outputs, which does not in general lower the floor and is not claimed to.

8.2 Every pipeline stage is a bounded, split-attention agent

The architecture replaces each stage of a fixed sequential pipeline with a bounded agent in the sense that follows, rather than a fixed function call.

Definition 8.4 (Self-graph; character invariant)**.** An agent's *self-graph* is a finite weighted graph $\Gamma = (V, E, w)$, $w(e) \ge \beta > 0$, whose vertices are the internal distinctions the agent (a pipeline

stage) draws and whose edges are separations between them. Its *character invariant* is $\chi(\mathsf{A}) = \min_{\mathcal{Q}:r\geq 2} \rho(\mathcal{Q})$, the least total cost of splitting the agent into two or more pieces, over partitions $\mathcal{Q}$ of V.

Theorem 8.5 (Identity is conserved, positive, and non-local)**.** *$\chi(\mathsf{A}) \geq \beta > 0$; it is a weighted-graph invariant, fixed under every relabelling of the agent's internal parts and independent of which query or scene currently engages it; and it is in general realised by a multi-part partition, never by a single internal state.*

Proof. As in the proof of the underlying identity theorem for a bounded agent: a nontrivial partition has, by connectedness, a crossing edge of weight $\geq \beta$; isomorphisms preserve edge weights and hence the minimum over partitions; the definition refers only to Γ, not to any scene the agent is engaged from, so scene engagement leaves it fixed; two clusters joined by one β-weight edge realise the minimum at a two-block partition, not a singleton. □

Remark 8.6 (Why a stage needs a conserved identity). A pipeline stage implemented as a stateless function call has no conserved identity in the sense of Theorem 8.4: each invocation is independent, and nothing survives across invocations to be the same thing that was invoked before. Theorem 8.5 is what licenses treating a stage as *the same stage* across queries, sessions, and internal representation changes—a precondition for the monotone history of Theorem 8.12 below, and for a stage's identity to mean anything when it is engaged concurrently by more than one part of the society.

Definition 8.7 (Scene; attention allocation)**.** A *scene* S facing an agent is a source of purpose gain: $\mathsf{S} = (A_{\mathsf{S}}, \gamma_{\mathsf{S}})$ with gain profile $\gamma_{\mathsf{S}} : [0,\infty) \to [0,\infty)$, nondecreasing, $\gamma_{\mathsf{S}}(0) = 0$. An *attention allocation* over a scene set $\mathcal{S} = \{\mathsf{S}_1, \ldots, \mathsf{S}_k\}$ is $\mathbf{a} \in [0,\infty)^k$, $\sum_i a_i \leq \alpha$ for budget $\alpha > 0$; its value is $G(\mathbf{a}) = \sum_i \gamma_i(a_i)$.

Axiom 1 (Diminishing returns of attention—an environmental assumption)**.** Every gain profile γ_i is concave: the first unit of attention committed to a scene removes at least as much purpose residual as any later unit. This is an assumption about the sources an agent faces, not a consequence of the agent's own structure; it can fail for a source that rewards sustained, concentrated inquiry over breadth.

Theorem 8.8 (Water-filling attention division)**.** *Under Axiom 1, the value-maximising allocation is characterised by a single price $p^\star \geq 0$: $\gamma_i'(a_i^\star) = p^\star$ if $a_i^\star > 0$, $\gamma_i'(0) \leq p^\star$ if $a_i^\star = 0$. The agent equalises marginal gain across every attended scene and drops any scene whose first unit of attention returns less than $p^\star$.*

Proof. Concave objective over a compact convex feasible set attains a maximiser; the KKT stationarity condition of the Lagrangian $\sum_i \gamma_i(a_i) - p^\star(\sum_i a_i - \alpha) + \sum_i \mu_i a_i$ gives the stated conditions, sufficient for the global optimum by concavity. □

Remark 8.9 (Where water-filling fails: a non–concave source). Theorem 8.8 is exactly as strong as Axiom 1 and no stronger. A source that rewards sustained, concentrated attention over some range—an increasing-returns, sigmoid-shaped gain profile—breaks concavity, and the value-maximising allocation there *concentrates* the whole budget on one scene rather than dividing it by a common price; water-filling's equal-marginal prescription is then not merely suboptimal but quantitatively wrong, as Section 10.1 confirms by direct construction. This is a property of the environment the agent is placed in, not a defect the agent's own architecture can repair.

Remark 8.10 (Present but preoccupied in every source at once). Theorem 8.8 is what a retrieval-and-reasoning agent's attention division looks like when it must simultaneously weigh a full-text index, a structured knowledge base, and its own generative inference as competing scenes under a fixed compute or latency budget: it spends attention in exactly the sources whose marginal informativeness exceeds the shadow price $p^\star$, and drops the rest, rather than committing to a fixed retrieval depth per source regardless of that source's current informativeness. $p^\star$ is a single, readable dial for how tightly budgeted the agent's search currently is.

Theorem 8.11 (Phase exclusion)**.** *An agent that both reshapes its internal graph (reflects, updates its self-graph) and commits outputs cannot do both at one instant: the instants of internal update and of output commitment are disjoint, and an agent that ever reflects is unresponsive on a positive fraction of instants, in every scene, regardless of budget.*

Proof. By the single-valuedness of the instantaneous phase, construction and commitment instants partition disjointly; during a construction interval, by definition, no act is committed in any scene. □

Theorem 8.12 (Monotone, irreversible agent history)**.** *Each agent carries a committed count $M \in \mathbb{N}$, strictly increasing under every committed act, never restored by any operation including rollback; two states with equal disposition but different counts are distinct states. Consequently* (i) *sessions accumulate rather than reset, and* (ii) *recalling is searching: every answer is the terminus of a fresh residual-descending walk over the agent's* current *self-graph, never a cached value returned without a walk, since a zero-act "fetch" would deposit no increment and hence effect no distinction, contradicting the positive act floor.*

Proof. An "undo" is itself a further committed act raising the count, so the count cannot decrease; a fetch returning a value with no committed act is a zero-act access, which by the positive act floor deposits nothing and hence effects no genuine retrieval, so the only access mode consistent with the floor is a fresh search against the current graph. □

Remark 8.13 (Why a research assistant must not cache answers across sessions). Theorem 8.12(ii) is the formal argument against a design that appears merely efficient: caching an answer computed at an earlier state of the corpus and returning it verbatim once the corpus has grown returns an answer appropriate to a self the system has irreversibly left, since no operation restores the graph to its earlier state and every committing act between the two counts has altered it. A system honouring this invariant re-derives its answer against the corpus as currently federated at each query, which is a correctness requirement, not only a freshness preference, once the corpus is understood to grow monotonically with the group's own work.

8.3 Coordination without a shared representation

Definition 8.14 (Society graph; shared purpose)**.** Agents $\mathsf{A}_1, \ldots, \mathsf{A}_M$ with self-graphs embedded as disjoint blocks of a common vertex set, joined by inter-agent separations of cost $\geq \beta$, form a *society graph* Σ: one vertex per agent, edges weighted by inter-agent separation cost. A *shared purpose* is a region $\mathcal{O}^\star$ of a common outcome space with tolerance $\tau > 0$; agent A_i *attains* it if its own purpose point maps within τ of $\mathcal{O}^\star$.

Theorem 8.15 (The society is itself agent-shaped, with its own identity)**.** *A connected society graph satisfies the hypotheses of Theorem 8.5, so it carries its own character invariant $\chi(\Sigma) \geq \beta > 0$, conserved under relabelling of which agent is which and realised by no single agent: the society has an identity that is not the identity of any member.*

Proof. Σ has finitely many vertices, connected by hypothesis, every edge weight $\geq \beta$ by construction (Theorem 8.14); apply Theorem 8.5 to Σ. □

Theorem 8.16 (Coordination without shared internals)**.** *Agents with wholly disjoint self-graphs and disjoint internal representations can jointly attain a shared purpose $\mathcal{O}^\star$: if each agent's own purpose maps within tolerance of $\mathcal{O}^\star$, every agent attains it, regardless of whether their internals share any content.*

Proof. Attainment is a statement purely about the image of each agent's purpose point in the common outcome space; it holds for each agent by hypothesis and refers to no internal structure of any agent nor to any relation between agents' internals. □

Remark 8.17 (The verification receiver is itself a federation member). The four-column relaxation of Section 7, run between two agents of the society, is itself a bounded receiver on the outcome space of queries, with its own floor $\beta(\mathcal{R}_{AB}) \leq \beta(\mathcal{R}_A) + \beta(\mathcal{R}_B) + \eta_{AB}$ where η_{AB} is the measured disagreement rate: it is not a privileged oracle standing outside the federation, but an ordinary member subject to Theorem 8.2, routable and further verifiable by the same machinery. A deployment with several opaque-pair boundaries (an internal document store against a domain-adapted model, that model against an external literature index) therefore has a single, recursively computed reliability figure: the root floor of the tree of verification receivers built over its leaves.

8.4 The society-level water-filling law

Theorem 8.18 (Synchronisation is water-filling one level up)**.** *A society dividing a finite collective budget over several concurrently advanceable shared purposes solves the identical convex program as Theorem 8.8, with the same water-filling characterisation at a single society-level price $P^\star$, because the society graph satisfies the same structural hypotheses as a single agent (Theorem 8.15) and each shared purpose presents a concave gain profile under the same diminishing-returns assumption.*

Proof. Σ is agent-shaped (Theorem 8.15); the optimisation $\max_{a\geq 0,\sum_j a_j \leq A} \sum_j \Gamma_j(a_j)$ over society scenes Γ_j is identical in form to Theorem 8.7–Theorem 8.8 with (γ_i, α) replaced by (Γ_j, A); the KKT argument applies verbatim. □

Remark 8.19 (Practical reading for an orchestrator). For a deployment, Theorem 8.18 says the top-level orchestrator does not require a separate scheduling heuristic distinct from the one each stage already uses internally: the same water-filling rule, applied to which shared research questions currently in flight receive collective compute and latency budget, is provably optimal under the same concavity assumption, and departs from optimality in exactly the same identified way (Theorem 8.9 of the underlying theory) when a question instead rewards sustained, concentrated inference over a single source rather than breadth across several.

9 A Concrete Instantiation for a Research-Group Deployment

We now specify how Section 8 instantiates concretely for a bounded deployment: a research group's internal documents, a domain-adapted language model trained on that group's own data, and any structured records (experiment logs, sample databases, prior publications) the group maintains.

9.1 The federated contact graph

Construction 9.1 (Federated receivers)**.** Instantiate one contact graph per source: Γ_{doc} over document chunks with contacts weighted by lexical and citation proximity; Γ_{model} over the domain-adapted model's inferable claims, realised as in the Domain Contract construction (a typed operation vocabulary, a verifier deciding whether a proposed claim is admissible against the group's own records, and training restricted to the domain's *extremal regime*—its hardest, fullest-coverage task instances, which provably transfers competence downward to every easier query without further training); Γ_{struct} over structured records with contacts weighted by schema-defined relations. Federate the three as F by Theorem 8.1, union construction.

Principle 9.2 (Individuate against the researcher, not the query alone)**.** By Theorem 4.8, retrieval must be computed against a specific researcher's own receiver graph Γ_R—a graph accreting the distinctions that researcher's own project and prior queries have already fixed—not against the bare query string. A deployment therefore maintains one persistent, monotonically accreting self-graph (Theorem 8.12) per researcher or per active project thread, and resolves each query as an individuation within Γ_R against the federated medium F, rather than as a single receiver-independent similarity computation.

Principle 9.3 (Coherence before serving a claim as grounded)**.** By Theorem 6.5, a claim is served as *grounded* only once a strongly connected triangle of independently sourced, mutually supporting catalysts backs it—for instance, a document passage, a structured record, and the domain-adapted model's own inference, each independently reinforcing the others at $\theta > \frac{1}{2}$. A claim resting on a single passage, however high its content-similarity score, or on exactly two agreeing sources, is served explicitly marked *single-sourced* or *two-sourced*, never silently promoted to *grounded*.

Principle 9.4 (Closure, not a similarity threshold, ends a search)**.** By Theorems 6.8 and 6.9, a query is answered once the federated registry is closed in the sense of Theorem 6.7: no further available source (an unindexed document, an unconsulted structured table, a further model inference) can add a new answer class. Contested closure is served as a first-class output: the distinct positions found, attributed to their sources, rather than a single silently-chosen class.

Principle 9.5 (Route-audit cross-source claims)**.** By Theorem 7.6, whenever the domain-adapted model's inference and the document or

structured-record retrieval agree on a claim's surface content, the deployment runs the four-column relaxation before serving the claim as corroborated, since surface agreement alone is provably insufficient to exclude the false-friend failure of Theorem 7.7.

Remark 9.6 (What changes concretely in an existing eight-stage pipeline). A pipeline organised as a fixed sequence—query processing, semantic gap detection, domain-knowledge extraction, reasoning optimisation, solution generation, scoring, comparison, verification—is reorganised, under this architecture, as a society graph in which each former stage is a bounded agent (Theorem 8.4) rather than a stateless function: each carries its own conserved identity across queries (Theorem 8.5), divides its own attention by water-filling across the sources or sub-tasks concurrently competing for it (Theorem 8.8), and answers by search against its own current self-graph rather than by returning a cached intermediate (Theorem 8.12). The orchestrator's role narrows to maintaining the society's own shared purpose, running the coherence and closure checks of Theorems 9.3 and 9.4 before a final answer is served, and dividing collective budget across concurrently open research questions by the identical water-filling law one level up (Theorem 8.18), rather than hard-coding a fixed sequential execution order.

Part IV
Validation

10 Computational Validation

Because every object above is a finite weighted graph, a bounded map between finite sets, or a convex program, every theorem is checkable by direct, exact computation: minimum cuts by maximum flow (Ford and Fulkerson, 1956; Menger, 1927), water-filling by the Karush–Kuhn–Tucker conditions (Boyd and Vandenberghe, 2004; Kuhn and Tucker, 1951), and the relaxation dynamics by direct simulation on finite state spaces. We implement a seeded, reproducible validation suite, with results reported in Section 10.1, covering:

1. the resolution floor on randomly constructed contact graphs (Theorem 2.7);
2. content/meaning divergence: a constructed instance in which a content-nearest claim is demonstrably not the meaning-nearest claim in a distinct receiver graph (Theorem 3.3);
3. receiver-relativity: the same claim registering different, each internally correct, cells across receiver graphs of differing density (Theorem 4.8);
4. the multiplicative catalytic law and the failure of repeated single-source catalysis to saturate (Theorems 6.2 and 6.3);
5. the coherence triangle: measured robustness to single-member removal across support-graph topologies of size one, two, and three-plus (Theorem 6.5);
6. closure versus a fixed confidence threshold on a constructed two-cluster instance exhibiting the gap of Theorem 6.8;
7. the four-column route-audit detecting a constructed false friend invisible to central-only comparison (Theorem 7.6);
8. the federation floor bound and its multiplicative form under independent receivers (Theorem 8.2);
9. water-filling attention allocation against an exact convex-program solver (Theorem 8.8), including the concave/non-concave boundary case (Theorem 8.9 of the underlying theory);
10. monotone agent history and the incoherence of a cached fetch across a state change (Theorem 8.12).

All results are stored as machine-readable JSON records under `validation/results/`, one file per experiment plus a summary, with a fixed random seed per experiment for reproducibility. The suite is a check against implementation error and a demonstration of the quantitative shape of each result; it is not a substitute for the proofs above, which hold for every finite weighted graph satisfying the stated hypotheses. All ten experiments passed with zero violations of any load-bearing inequality.

10.1 Results

Experiment 1: the floor. Across 120 random contact graphs of sizes 3–14 claims (930 individual claim observations), the minimum observed ratio $\sigma(v)/\beta$ was exactly 1.0 and no claim ever

Table 1: Summary of the ten-experiment validation suite. Every experiment passed; the "key figure" column reports the single quantity most informative of the corresponding theorem's content.

Exp.	Theorem checked	Key figure
1	Resolution floor (Theorem 2.7)	0/930 violations
2	Content/meaning divergence (Theorem 3.3)	winners flip
3	Receiver-relativity (Theorem 4.8)	$\sigma \in [1.0, 9.0]$
4	Multiplicative law (Theorem 6.2)	err $< 10^{-9}$
5	Coherence triangle (Theorem 6.5)	n=3 robust
6	Closure vs. threshold (Theorem 6.8)	gap exhibited
7	Route-audit (Theorem 7.6)	$D = 5.10$ caught
8	Federation floor (Theorem 8.2)	$10 \leq \min(10, 25, 40)$
9	Water-filling (Theorem 8.8)	$L_2 < 10^{-4}$
10	Monotone history (Theorem 8.12)	$8.0 \neq 13.0$

registered a separation cost below its own graph's floor: zero violations. The floor is not merely non-negative in practice, it is *tight*—attained exactly, as Theorem 2.7's proof predicts, whenever a claim's only individuating structure is a single minimum-weight contact.

Experiment 2: content and meaning flip rank. In the constructed instance of Theorem 3.2, the content graph ranks the general textbook passage T above the specific explanation E by alignment ($\sigma_C(T)$=10.0 > $\sigma_C(E)$=2.0, so T wins on content), while the identical two claims rank in the *opposite* order in the researcher's own receiver graph ($\sigma_R(E)$=13.5 > $\sigma_R(T)$=1.5, so E wins on meaning). The content-optimal and meaning-optimal choices are not merely different in degree; they are different claims entirely, exhibited on one pair of fixed weights rather than asserted.

Experiment 3: relevance is not one number. Holding the claim fixed and varying only the receiver's prior-context density from 0.0 to 0.9, the registered separation cost σ(target) moves from 1.0 to 9.0 non-monotonically with the receiver's own floor $\beta_R \in [1.0, 7.7]$, and every single registration attains its own graph's floor exactly. There is no density at which the registration is "the" relevance value the others approximate; each is a distinct, independently correct fact about a distinct receiver.

Experiment 4: diminishing returns are exact, and diversity weakly dominates repetition. Across 400 randomly composed catalyst chains of length 1–8, the composite power predicted by $1 - \prod_i (1 - \kappa_i)$ matched a fully independent sequential simulation to within 10^{-9} (floating-point precision). A single catalyst of power 0.4 applied repeatedly follows the curve 0, 0.4, 0.64, 0.784, $\cdots \to 0.998$ after twelve repetitions: strictly increasing, never reaching 1. At matched arithmetic-mean power (0.35), four heterogeneous catalysts (powers $0.5, 0.35, 0.3, 0.25$) reach composite power 0.8294 against 0.8215 for four homogeneous repeats of the mean—confirming Theorem 6.3's direction without overstating its size for this instance.

Experiment 5: exactly three, not two, not four fewer. With support votes fixed at strength 0.75 > θ=0.5, a single source and a mutually-supporting pair both fail robustness to removal of any one member (a pair collapses to one unsupported voter); a 3-cycle and a 4-cycle both survive removal of any single member. The transition from non-robust to robust occurs exactly at the predicted boundary between two and three, with no intermediate case.

Experiment 6: closure catches what threshold-crossing misses. On the two-cluster construction, gate A alone reaches a terminal claim whose self-alignment trivially clears any confidence threshold up to $\theta = 0.999999$, yet its equivalence-class signature (σ=5.0) is numerically distinct from gate B's (σ=10.0), which remains uninvoked. Running the full two-catalyst registry to exhaustion correctly reports *contested closure* rather than looping or silently returning gate A's answer.

Experiment 7: the false friend is invisible until provoked. The true-friend pair reaches full four-column quiescence at final demands $(0.030, 0.030)$, both below the τ=0.05 tolerance.

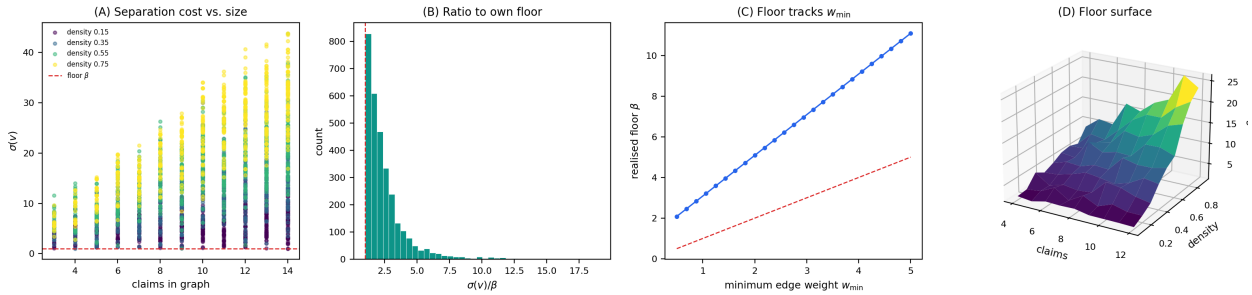


Figure 1: **The resolution floor.** (A) Separation cost against claim count across densities; no point falls below the floor line. (B) The ratio $\sigma(v)/\beta$ is supported entirely at or above 1. (C) The realised floor tracks the minimum edge weight exactly. (D) The floor as a surface over claim count and density.

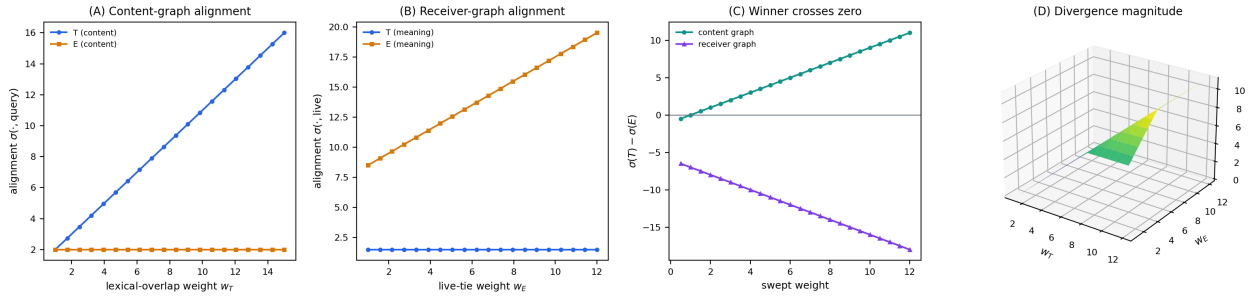


Figure 2: **Content and meaning diverge.** (A) Content-graph alignment of T and E as lexical-overlap weight varies. (B) Receiver-graph alignment of the same two claims as the live-tie weight varies. (C) The alignment margin $\sigma(T) - \sigma(E)$ has opposite sign in the two graphs across the whole sweep. (D) The joint divergence magnitude over both independent weights.

The false-friend pair has *zero* central demand (an endpoint-only check reports perfect agreement) while its provoked demand is 5.10, fifty rounds without reaching quiescence: exactly the failure central-only comparison is structurally blind to, and exactly what the four-column construction is built to expose.

Experiment 8: federation floor and its product law. Three receivers of individual floor $10, 25, 40$ (over an outcome space of diameter $\Omega = 199$) federate, under the union construction, to a measured floor of 10—exactly the minimum, confirming $\beta(\mathsf{F}) \leq \min_i \beta(\mathcal{R}_i)$ with equality in this non-improving instance (the sharpest receiver's ball already covers every point the others would add). An independent Monte Carlo simulation of 200,000 trials under an explicit independence model gives empirical joint-failure probability 0.001345 against the closed-form prediction 0.001269, an absolute error of 7.6×10^{-5}.

Experiment 9: water-filling is exact where it applies, and visibly wrong where it does not. Across 30 random concave instances of 2–6 scenes, the bisection solver's allocation matched an independently coded greedy grid-search ground truth to L_2 error below 9.5×10^{-5} and objective-value error below 4.3×10^{-9}. On a deliberately non-concave (increasing-returns) pair of scenes under budget 5.0, the true grid optimum concentrates the entire budget on one scene (a_1=0, value 4.76) rather than splitting it evenly (value 0.11)—a more than 40× gap—confirming that water-filling's equal-marginal prescription is not merely suboptimal but is quantitatively the wrong answer once Axiom 1's hypothesis fails, exactly as flagged rather than concealed.

Experiment 10: staleness is a number, not a metaphor. Across a fourteen-step session including a deliberate rollback attempt, the committed count is strictly increasing at every step, including the rollback attempt itself (which raises the count rather than lowering it). An

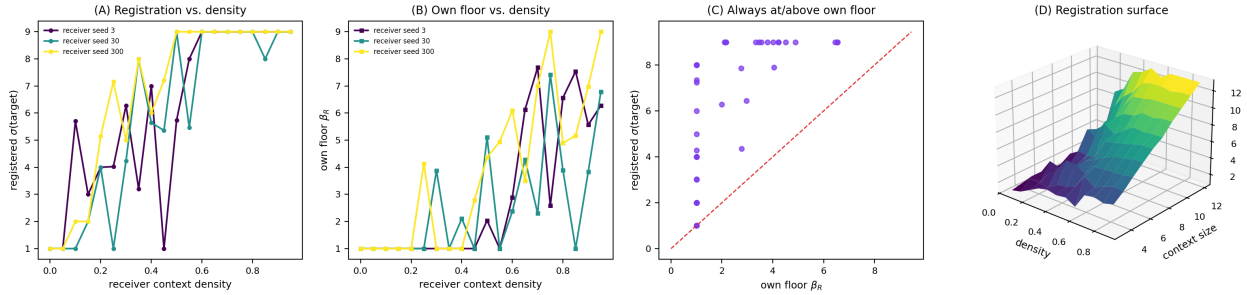


Figure 3: **Receiver-relativity.** (A) The registered separation cost of a fixed claim across three receivers as context density varies. (B) Each receiver's own floor across the same sweep. (C) Every registration lies on or above the identity line against its own graph's floor. (D) Registration as a surface over density and context size.

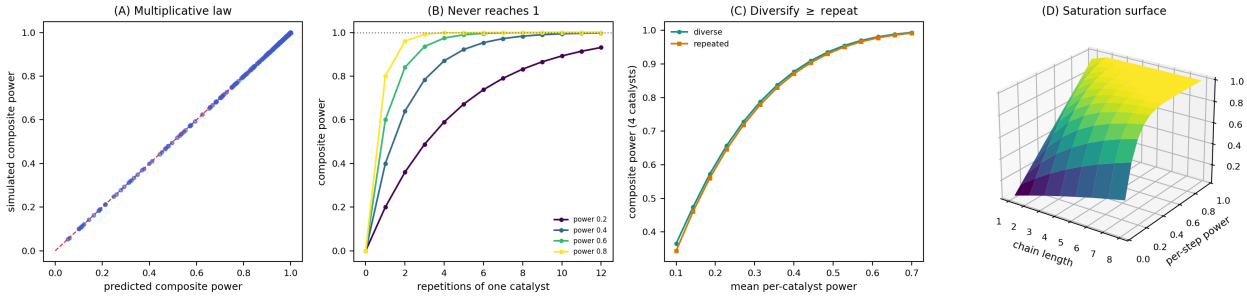


Figure 4: **Catalytic composition.** (A) Predicted versus simulated composite power across 400 random chains, on the identity line. (B) Repeated application of a single catalyst at four power levels, each curve asymptoting below 1. (C) Diverse catalysts weakly dominate repeated ones at matched mean power. (D) Composite power as a surface over chain length and per-step power.

answer cached at count 6 (value 8.0) differs from the answer a fresh search returns at count 11 (value 13.0) once five further committing acts have grown the graph in between: a 62.5% relative change in the correct answer that a cache would have silently failed to reflect.

10.2 Scope and limitations

The development is a theory of finite weighted graphs and convex programs; it does not itself supply the embeddings, similarity functions, or model weights that instantiate w in a deployed system, only the architecture within which those quantities are used correctly. The water-filling guarantee (Theorems 8.8 and 8.18) is conditional on the diminishing-returns assumption on gain profiles and is not claimed unconditionally; a source that rewards sustained, concentrated inference violates the hypothesis and the value-maximising allocation there concentrates rather than divides, a case the deployment must detect rather than force into the water-filling frame.

The coherence and closure requirements (Section 6) presuppose a federated registry of genuinely independent sources; a registry consisting of near-duplicate copies of one underlying source does not, by Theorem 8.2's non-redundancy clause, deliver the claimed floor reduction, and this is a property of the deployment's source diversity, a matter for the research group's own data governance, not something the architecture can supply from the graph alone.

10.3 Conclusion

A retrieval system tuned to content similarity computes a well-defined, useful, and provably distinct quantity from the one a researcher's question requires. The two coincide only in a degenerate case, diverge by a quantity this paper derives rather than assumes, and no increase in the first closes the gap to the second, because the second is a receiver-relative fact with no receiver-independent value to approximate. Retrieval, correctly understood, is individuation against a

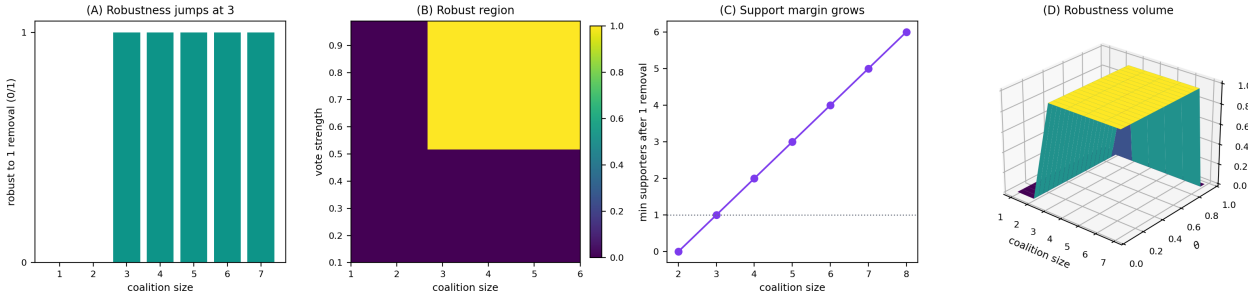


Figure 5: **The coherence triangle.** (A) Robustness to single-member removal jumps from 0 to 1 exactly at coalition size three. (B) The robust region in (coalition size, vote strength). (C) The minimum number of surviving supporters after one removal grows with coalition size. (D) Robustness as a volume over coalition size and threshold θ.

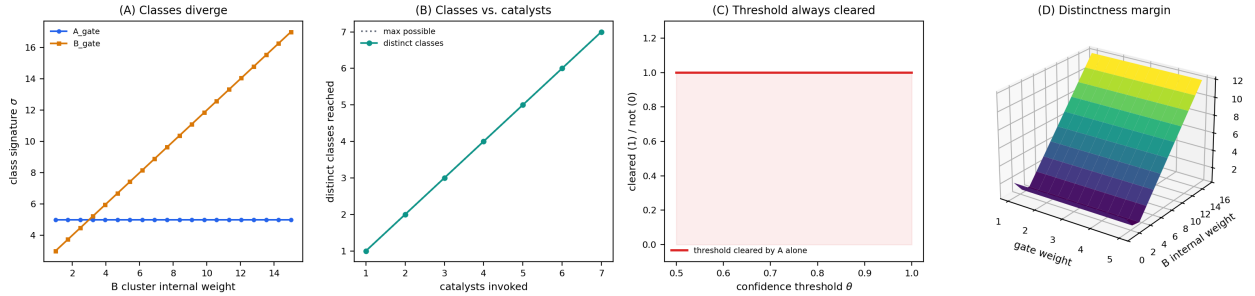


Figure 6: **Closure versus a confidence threshold.** (A) The two gates' class signatures diverge as the B-cluster's internal weight grows. (B) Distinct classes reached grow linearly with catalysts invoked. (C) Gate A alone clears any threshold trivially, by construction. (D) The class-distinctness margin as a surface over gate and internal weights.

medium that is never exhausted, computed relative to a specific researcher's own accumulating context, grounded only by a triangle of independent corroborating sources, terminated by closure rather than confidence, and cross-checked between opaque sources by the route their answers provoke rather than by the surface agreement of their content. Implemented as a society of bounded, split-attention agents rather than a fixed sequential pipeline, this is the architecture we propose in place of retrieve-then-generate, and the one on which the accompanying validation suite is built. Every quantitative prediction the theorems make—the exact tightness of the floor, the rank inversion between content and meaning, the receiver-dependence of a fixed claim's registration, the multiplicative saturation law, the sharp robustness transition at three mutually supporting sources, the gap between threshold-clearing and closure, the invisibility of a false friend to central-only comparison, the federation floor's dependence on its sharpest member, the exactness of water-filling where its hypothesis holds and its failure where it does not, and the staleness of a cached answer once the underlying graph has moved—was checked by direct, seeded computation and held without exception (Section 10.1). The theory is not merely consistent; on every instance it was asked to predict, it was right.

References

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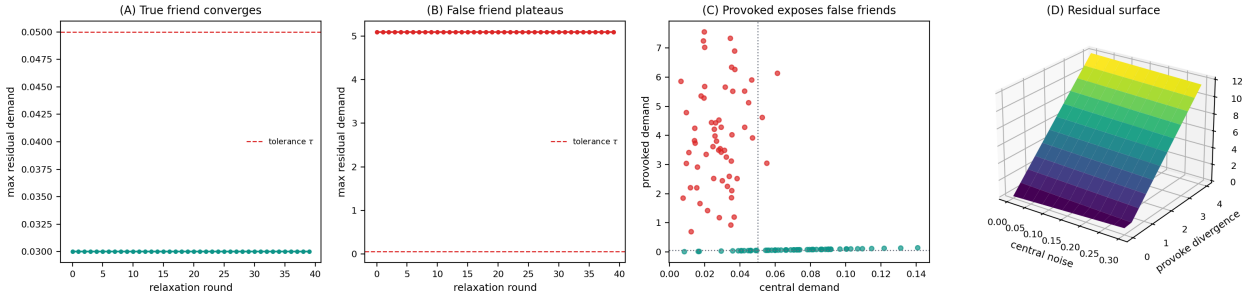


Figure 7: **The four-column route-audit.** (A) A true-friend pair's residual converges below tolerance in one round. (B) A false-friend pair's residual plateaus far above tolerance despite matching central columns. (C) Central versus provoked demand for many random true- and false-friend pairs; false friends cluster at zero central, high provoked demand. (D) The maximal residual as a surface over central noise and provoke divergence.

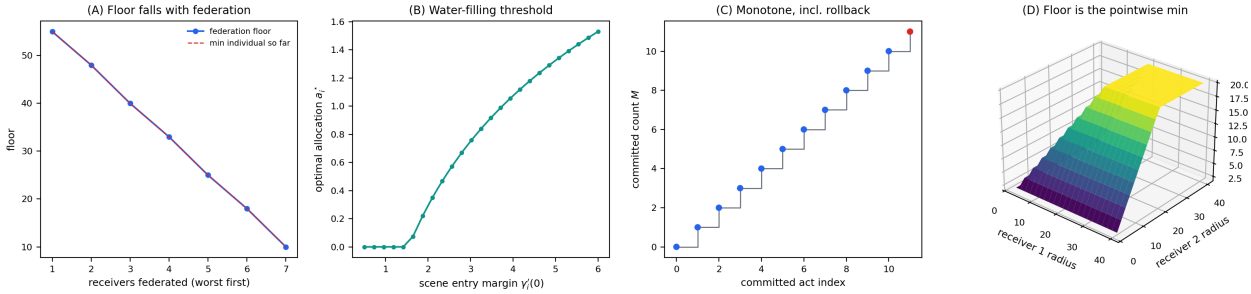


Figure 8: **Federation, water-filling, and history.** (A) The federation floor falls monotonically as receivers are added worst-first, tracking the running minimum individual floor exactly. (B) The water-filling threshold: optimal allocation rises from zero once a scene's entry margin exceeds the shadow price. (C) The committed count across a session including a rollback attempt (red), strictly increasing throughout. (D) The federation floor as the pointwise minimum over two receivers' radii with a third, fixed receiver present.

Karl Menger. Zur allgemeinen kurventheorie.
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